INSTRUCTIONS TO THE CANDIDATES

- Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

  Non-adherence with this instruction shall result in candidate being booked under malpractice.
- Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- Candidate should write only question number in the margin.
- Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- Invigilators should ensure that the above instructions are strictly followed by the candidates.
- After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- Do not crumple or cut the sides of the sheets.
- Do not write, mark or damage the timing tracks or Barcode.

3

LPP - Graphical Method.

Max $z = 3x + 4y$.

Subject to Constraints.

$x + 2y \leq 4$ — ①

$3x + 2y \leq 6$ — ②

$x \geq 0, \; y \geq 0$

eqn ① $x + 2y = 4$

when $x = 0$

$2y = 4$

$y = 2$.

When $y = 0$

$x = 4$

$x = 4$

x	0	4
y	2	0

eqn ② $3x + 2y = 6$.

when $x = 0$

$2y = 6$

$y = 3$

when $y = 0$

$3x = 6$

$x = 2$.

x	0	2.
y	3	0

Points for eqn ①

$x + 2y \leq 4$

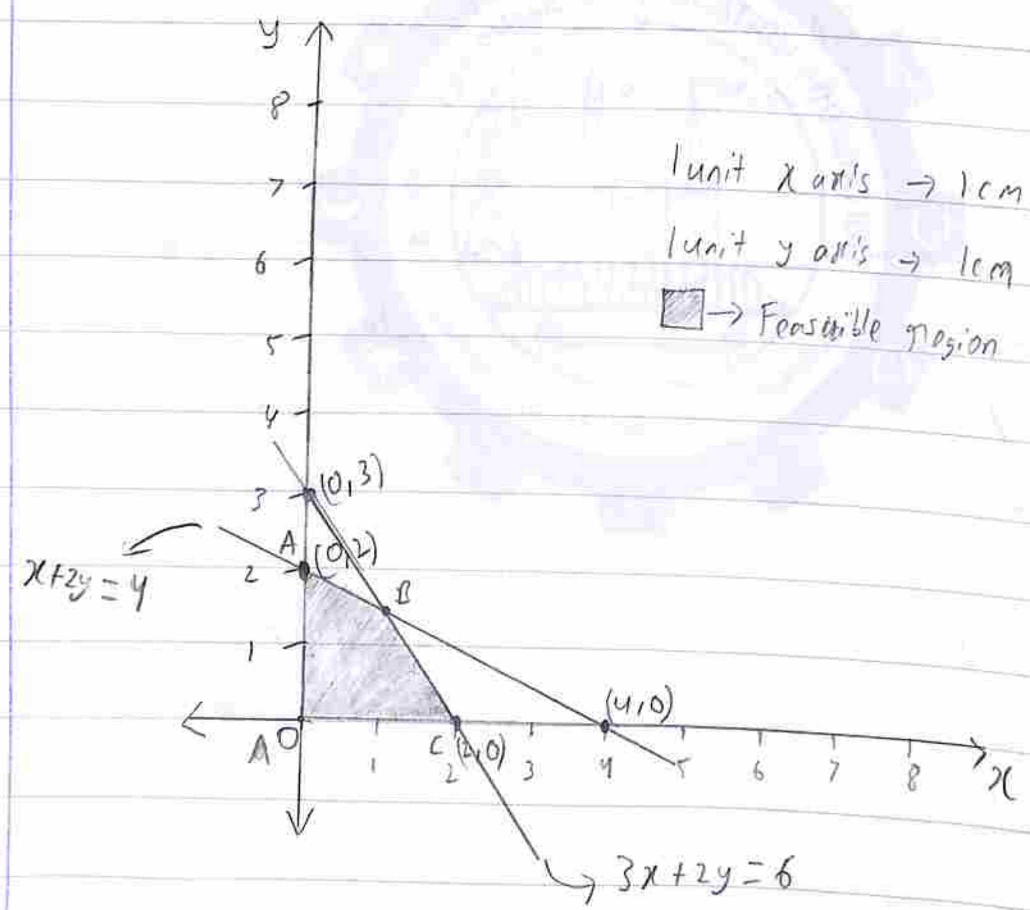
P

P (0,2) P(4,0)

Points for eqn ②

$3x + 2y \leq 6$

P(0,3) P(2,0)



Points of Feasiable region

O (0,0)

A (0,2)

C (2,0)

B (   ) → Solve using equations.

$x + 2y = 4$

$3x + 2y = 6$

$\Rightarrow \cancel{3x} + 6y = 12$

$(-)3x \, (-)+ 2y = (-)6$

$0 + 4y = 6$

$\boxed{y = \frac{6}{4}}$

$x + 2\left(\frac{6}{4}\right) = 4$

$x = 4 - \frac{12}{4}$

$\boxed{x = 1}$

$\therefore B(1, 6/4)$

$$Z = 3x + 4y$$

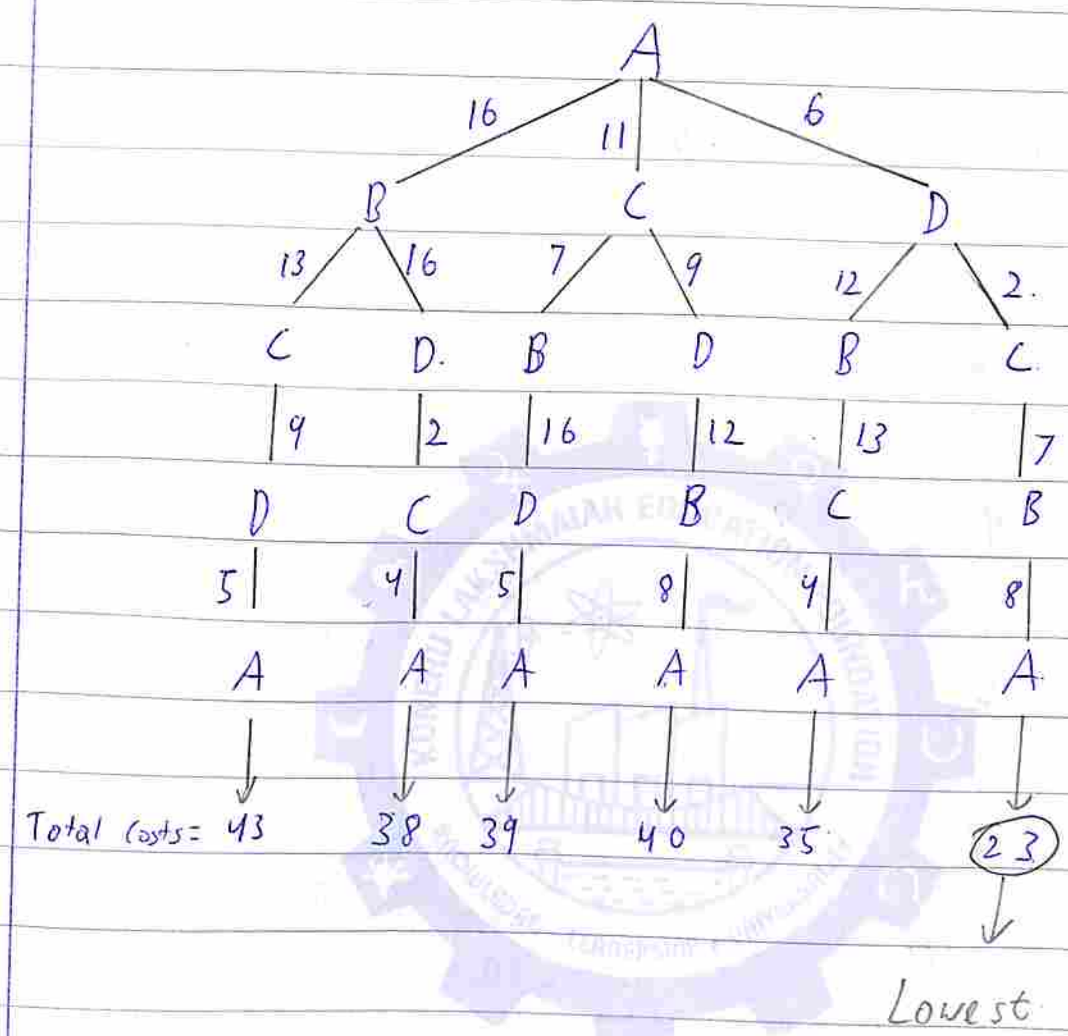
$O(0,0)$	$Z = 3(0) + 4(0)$	$= 0$
$A(0,2)$	$Z = 3(0) + 4(2)$	$= 8$
$B(1, 6/4)$	$Z = 3(1) + 4(6/4)$	$= 9$ Max
$C(2,0)$	$Z = 3(2) + 4(0)$	$= 6$

$\therefore$ $Z = 9$ when $\boxed{x = 1}$ and $\boxed{y = 6/4}$

5. Traveling Salesman.

	A	B	C	D
A	0	16	11	6
B	8	0	13	16
C	4	7	0	9
D	5	12	2	0

Cities A, B, C, D and cost details represented by the above cost matrix



The total minimum cost for traveling the cities is $\boxed{23}$

Order $\rightarrow$ A – D – C – B – A

2

2.A Transportation Problem - least cost method

	D_1	D_2	D_3	Supply
O_1	9	8	5	25
O_2	6	8	$\boxed{35}$ 4	35-35 =0
O_3	7	6	9	40
Demand	30	25	45-35 =10	100 / 100

	D_1	D_2	D_3	Supply
O_1	9	8	$\boxed{10}$ 5	=15 ~~25~~
O_2	6	8	$\boxed{35}$ 4	0
O_3	7	6	9	40
Demand	30	25	10-10 =0	

	D_1	D_2	D_3	Supply
O_1	9	8	$\boxed{10}$ 5	15
O_2	6	8	$\boxed{35}$ 4	0
O_3	7	$\boxed{25}$ 6	9	40-25 =15
Demand	30	25-25 =0	0	

	D_1	D_2	D_3	Supply
O_1	9	8	$\boxed{10}$ 5	15
O_2	6	8	$\boxed{35}$ 4	0
O_3	$\boxed{15}$ 7	$\boxed{25}$ 6	9	15-15 =0
Demand	30-15 =15	0	0	

	D_1	D_2	D_3	Supply
O_1	$\boxed{15}$ 9	8	$\boxed{10}$ 5	$15-15=0$
O_2	6	8	$\boxed{35}$ 4	0
O_3	$\boxed{15}$ 7	$\boxed{25}$ 6	9	0
Demand	$15-15=0$	0	0	

$$4(35) + 5(10) + 6(25) + 7(15) + 9(15)$$

$$140 + 50 + 150 + 105 + 135 = \boxed{580}$$

2.B Duality

$$\text{Max } Z = 20x_1 + 35x_2 + 49x_3$$

ST

$$2x_1 + 3x_2 + 4x_3 \le 60$$
$$9x_1 + 3x_2 + 6x_3 \le 96$$
$$2x_1 + 7x_2 + 8x_3 \le 70$$
$$x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0$$

~~ale~~ dual variables = $Y_1 \quad Y_2 \quad Y_3$

matrix form

$$\begin{bmatrix} 2 & 3 & 4 \\ 9 & 3 & 6 \\ 2 & 7 & 8 \end{bmatrix} \qquad \begin{bmatrix} 60 \\ 96 \\ 70 \end{bmatrix}$$

$$\text{Minimise } Z^* = 60Y_1 + 96Y_2 + 70Y_3$$

Minimise $Z^* = 60y_1 + 96y_2 + 70y_3$

Subject to

$$2y_1 + 9y_2 + 2y_3 \geq 20$$
$$3y_1 + 3y_2 + 7y_3 \geq 35$$
$$4y_1 + 6y_2 + 8y_3 \geq 49$$

2.C. Branch and bound

This Branch and bound technique is used with graphical method to Obtain Integer Linear programming

Advantges

* Obtains Integer value.

Disadvantages

* Time consuming

1

1. A Transportation Problem.

It is a problem about transportation of supply and demand. There are multiple ways to solve this.

* North west method
* least cost method
* vogel's method.
* row minima
* column minima

1. E Integer Linear Programming Problem.

Integer linear programming problem is about getting the final answer in Integer form. There a a few ways to acheive this

* Gomory's cut plane method
* Branch and bound method

It usally consists of adding additional constraints acquired from the problem to achieve Integer answer.

1.F Approximation Algorithm

This Algorithm is used on graphs.

IT is is used to find which are the most important nodes and ~~vectors~~ edges of the particular graph,

It provides a vector of nodes denoting their importance

1.D Travelling Salesman algorithm.

This Algorithm is used to compute the total minimum cost required by the Salesman to traverse.

The constraints are that if the salesman moves forward it can not go back to the nod it has visted except the starting node at the end after visting all the nodes

1.B Simplex method

The pivot element is the inter junction of the pivot row and pivot column.

1.

1.B

KONERU LAKSHMAIAH EDUCATION FOUNDATION
KNOWLEDGE · LEADERSHIP · UNIVERSITY